

Newcastle University Investment Fund

Universal Display Company – NASDAQ - (OLED)

Long: Target Price \$120.12 – (28.55% Upside)

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UNIVERSAL DISPLAY
C O R P O R A T I O N



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Executive Summary

Pitch Narrative

UDC owns the essential components inside every OLED screen. Its phosphorescent emitter materials and patents are embedded into the supply chains of Samsung, LG, and BOE. With smartphone OLED shipments hitting 890 million units in 2025 and a 76% gross margin model that scales with panel volume, UDC generates compounding royalty and materials revenue with no additional capex required.

Three catalysts converge in the same 12-month window: Gen 8.6 fabs from Samsung and BOE come online in mid-2026 producing IT panels that require 3-5x more material per panel than a smartphone; Apple's Tandem OLED rollout across iPad and MacBook effectively doubles revenue per device; and the new Tianma supply agreement opens a fast-growing Chinese customer base that was previously untapped.

LG Display has now verified UDC's blue phosphorescent material works at mass-production scale. Adoption is expected from 2026 which means UDC sells three emitter colours per panel instead of two, earns a new royalty layer, and triggers a global hardware refresh cycle, with a further ESG tailwind as regulators tighten standards on high-power display technology.

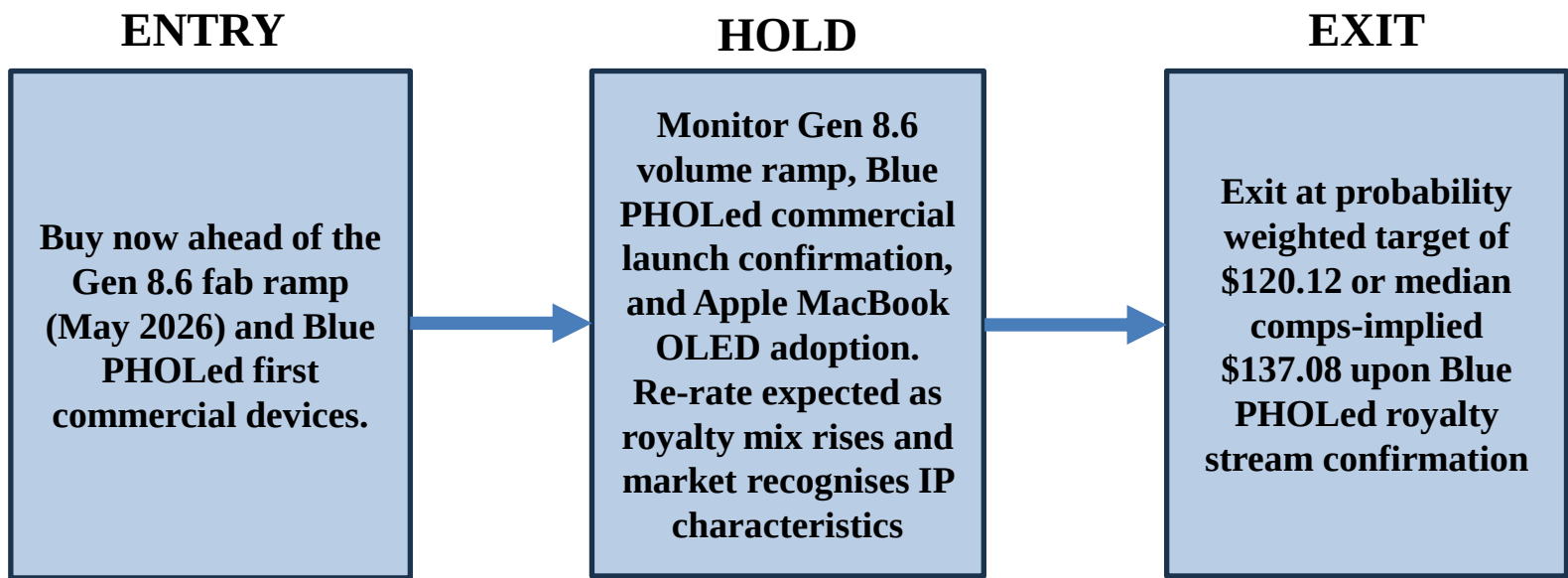
Street vs Variant Views

Street View	Variant View
UDC is a mature materials business with limited near-term revenue growth and FY25 guidance of \$650–700M is roughly flat, and the stock deserves a mid-cycle materials multiple.	The revenue mix is misunderstood. 76% gross margins and a recurring royalty stream make UDC structurally closer to an IP licensor like InterDigital (trading at 11x P/S) than a materials manufacturer.
Blue PHOLed is almost here and the market has heard this story for years and commercial adoption remains uncertain.	LG Display's mass-production verification is a genuine risk lowering event, not incremental news. Combined with Gen 8.6 fabs coming online in mid-2026, the revenue step has a credible 12-month timeline.
China expansion adds customer concentration risk given geopolitical uncertainty	The Tianma deal is structural demand diversification reducing dependence on Samsung and LG while opening a panel market growing faster than Korea

Valuation Summary

	Target Price	Upside	Weight	Weighted TP	Weighted Upside
BULL	\$147.32	57.66%	17.50%	\$120.12	28.55%
BASE	\$111.43	19.25%	35%		
BEAR	\$81.21	-13.09%	17.50%		
MEDIAN COMPS	\$137.08	46.70%	30%		

Entry and Exit Timeline



Company Overview

Company Overview

- ❑ Universal Display Corporation lead in production of OLED technologies and advanced materials used across various display applications.
- ❑ The company provides high-performance OLED panels for many of the world’s top electronics manufacturers.
- ❑ UDC develops key OLED technologies such as phosphorescent OLED (PHOLED) materials, alongside flexible and transparent, white OLED platforms that support more efficient displays.

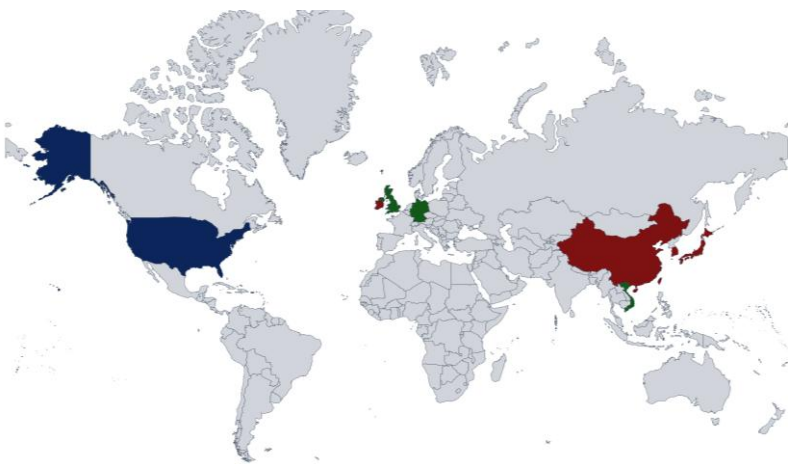
Competitive Advantages

- 1. PHOLED Technology**
 - Universal Display’s PHOLED technology is protected by a patent that underpins a high-margin licensing business model. This structure creates recurring revenue streams that scale as global OLED production expands.
- 2. Embedded in Supply Chain**
 - The company maintains multi-year supply and licensing agreements with Samsung, LG, BOE, and other manufacturers. These relationships provide stable demand and embed UDC’s materials into the core of the global display supply chain.
- 3. Pricing Power**
 - Decades of R&D and extensive patents make replication extremely difficult for competitors. This entrenched technological moat ensures UDC retains pricing power and long-term strategic relevance in the OLED ecosystem.

Management

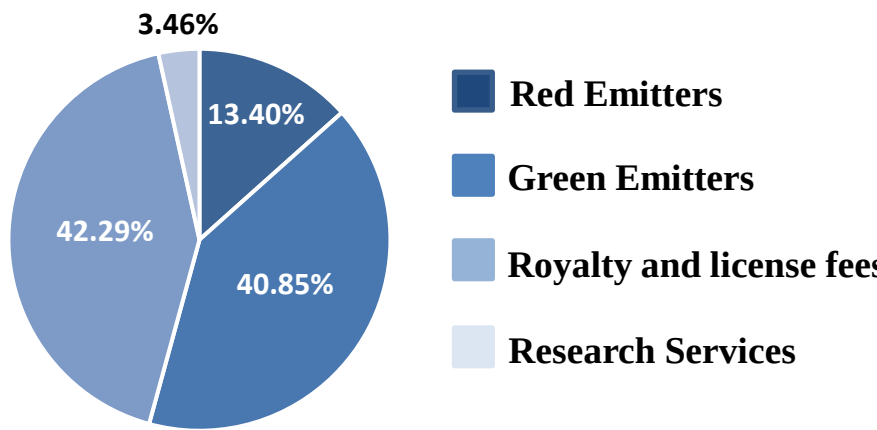
- Universal Display Corporation is led by an experienced team focused on advancing OLED materials science and expanding commercial adoption
- Leadership continues to prioritize Research and Development, IP expansion, and long-term supply partnerships with major global display manufacturers.
- CFO Brian Millard has highlighted the company’s strong financial momentum in 2024, reinforcing the scalability of UDC’s high-margin materials and licensing model.

Geography



- Universal Display's headquarters is situated in Ewing, New Jersey, United States.
- The company supports presence and customer relationships in China, South Korea, Japan, Taiwan, Ireland, and Hong Kong.
- It also collaborates with partners in Germany, Singapore, the United Kingdom, and Vietnam, reinforcing its reach across the OLED ecosystem.

FY25 Revenue breakdown per sector



Financials

Financial Metric	2025 Q4	2024 Q4
Total Revenue	\$172.93 million	\$162.29 million
Free Cash-Flow	\$15.06 million	\$22.04 million
Net Income	\$66.34 million	\$46.02 million
EPS (Diluted)	\$1.39	\$0.96



Industry Overview

Market Snapshot:

Year	Market Size	Growth (%)
2023	\$527bn	-8.2%
2024	\$628bn	+16%
2025	\$792bn	+20.7%

Industry Themes:

Highly capital-intensive: Complex, vertically integrated structure with major firms controlling fabrication, design, and equipment supply chains.

Technology-driven: Constant innovation in chip architecture, manufacturing processes, and performance capabilities drives competitive advantage.

Globalised with high barriers: Worldwide demand supported by large-scale fabs; oligopolistic structure with immense capital requirements limiting new entrants.

Cyclical but structurally growing: 2023's -8.2% contraction followed by 2024–25 rebound driven by AI, cloud, and advanced consumer electronics demand.

Structural shifts:

- Geopolitical fragmentation: Korea's strict technology transfer rules, US export controls on advanced chips, and EU/US CHIPS Acts are reshaping global supply chains and investment flows.
- AI & advanced node demand: AI computing, cloud infrastructure, and next-gen consumer electronics are driving sustained demand for cutting-edge semiconductor technologies.
- OLED's specialised position: OLED sits at the intersection of semiconductor manufacturing and display innovation, benefiting from rising demand for high-resolution, energy-efficient, flexible screens.

Key Customers

SAMSUNG

- World's largest OLED panel maker; dominant supplier to Apple iPhone.
- Leading Gen 8.6 fab investment for IT OLED ramp in 2026.
- Produces red & green PHOLED using UDC materials at mass scale.
- Revenue leadership in smartphone and foldable OLED segments.

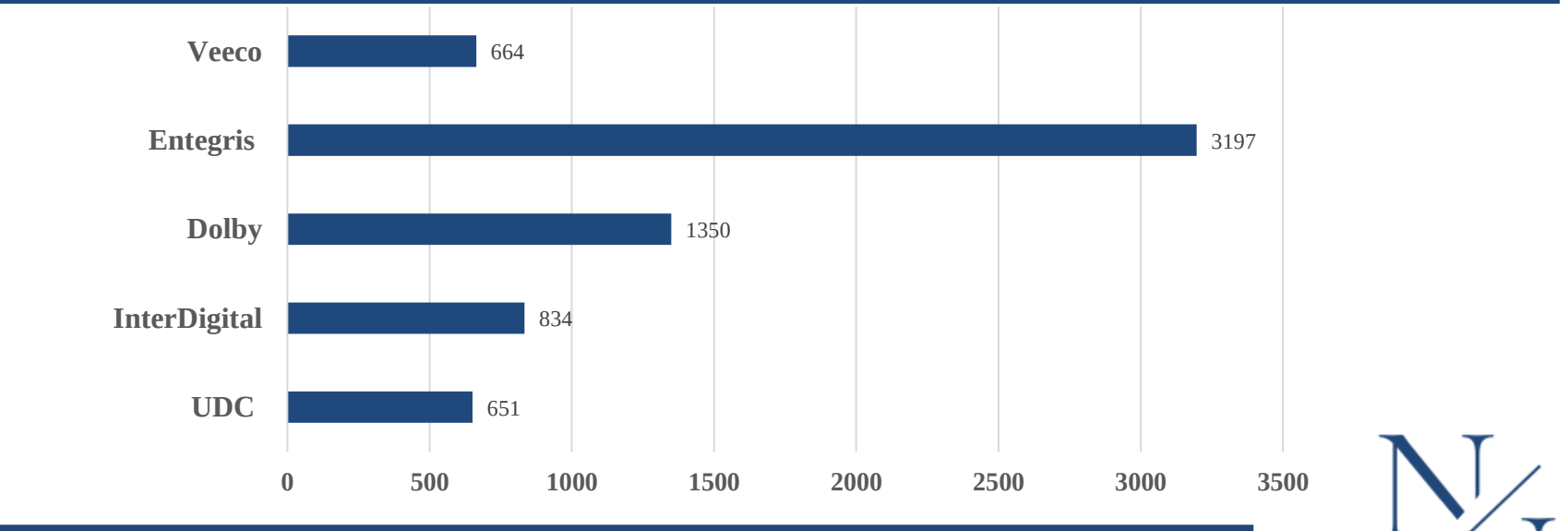
LG

- Global leader in large-area OLED; dominant in OLED TV panels.
- Verified UDC's blue PHOLED material works at mass-production scale.
- Targets 7 million OLED TV panels shipped in 2026.
- Exclusive supplier of wearable AMOLED displays to Apple.

BOE

- China's largest display manufacturer; fastest-growing OLED panel maker.
- Accelerating Gen 8.6 flexible AMOLED fab build-out for 2026..
- Key beneficiary of China's \$14.5bn OLED industry investment mandate.

FY2025 Revenue vs Peers (\$m)

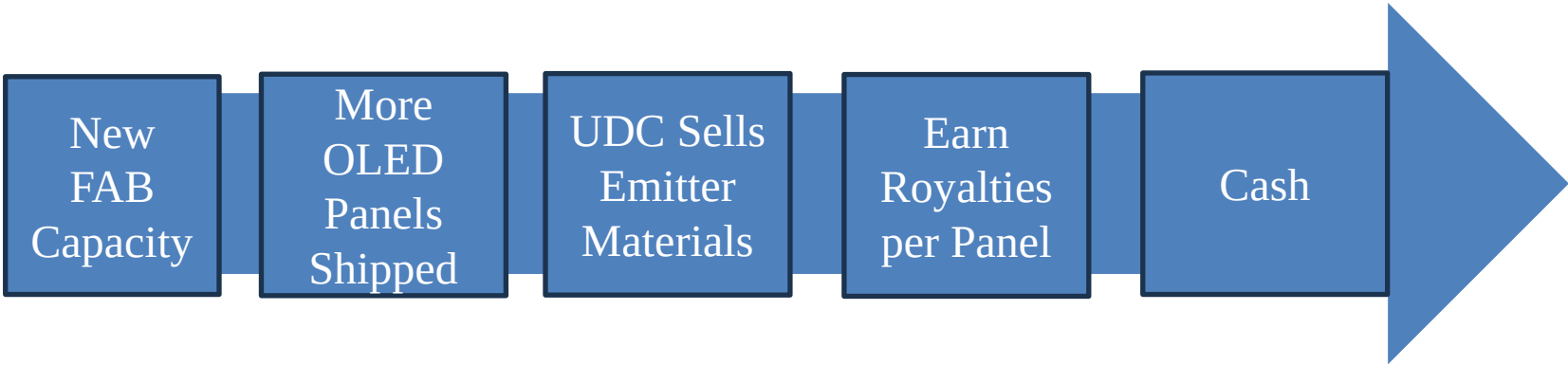


Thesis 1: OLED Market Expansion Driving Royalty and Revenue Growth

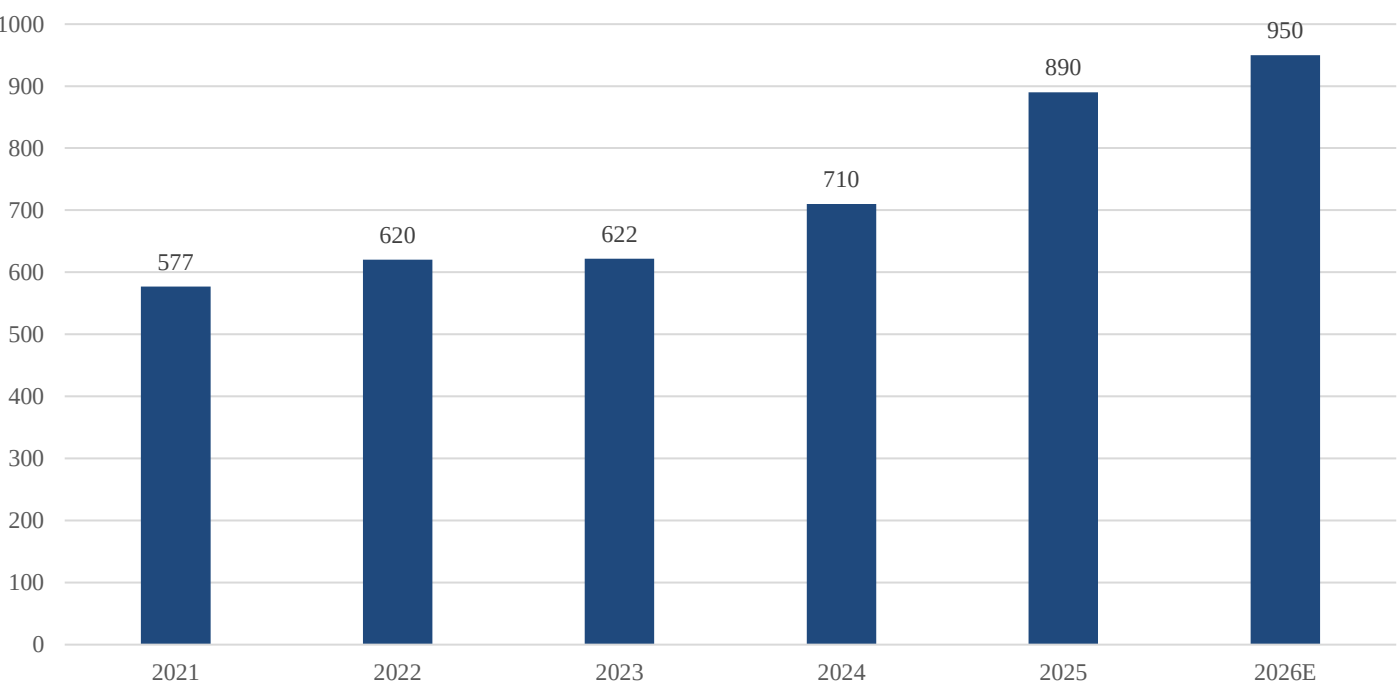
Structural Growth Drivers and Shareholder Returns

- **OLED and LCD**
Penetration expanding across laptops, tablets, monitors, and automotive displays. Gen 8.6 fabs coming online in 2026 mean significantly more panels, directly translating to higher royalty volumes and material sales for UDC.
- **Apple Tailwinds**
Apple's push into OLED for iPads and MacBooks pulls the entire upstream supply chain, with UDC's major customers (Samsung Display, LG Display) ramping capacity to meet Apple demand lifting royalty volumes
- **Chinese Growth Frontier**
UDC signed long-term supply agreements with Tianma, a leading Chinese panel maker, opening a fast-growing and previously underpenetrated customer base with significant long-run volume potential.
- **Shareholder Friendly**
Management raised the dividend by 11% in Q1 2026 and has been actively repurchasing shares — a strong signal of confidence in the durability of the earnings outlook.

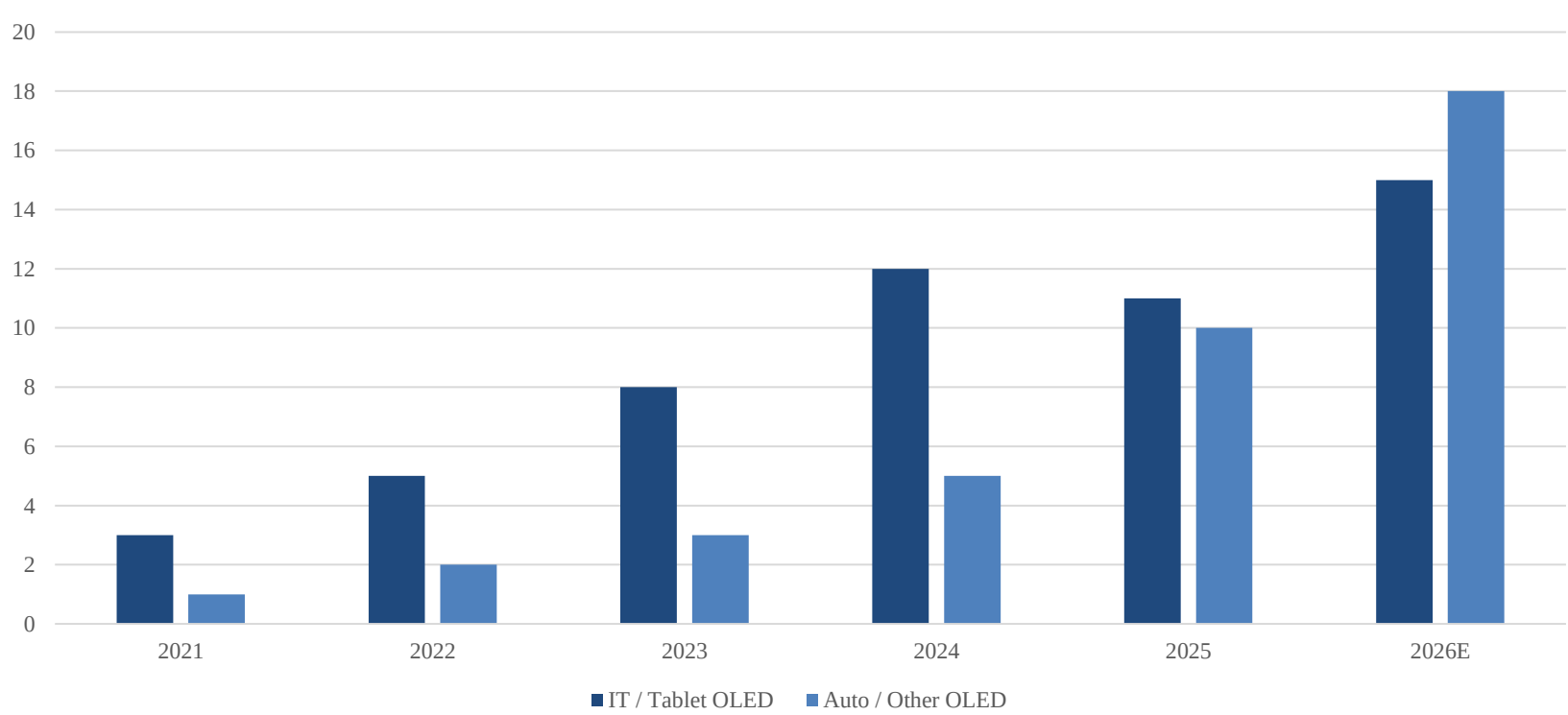
UDC Revenue Cycle



Smartphone OLED Shipments (millions)



IT/Tablet & Automotive OLED Shipments (millions)



Thesis 2: Blue PHOLED

The Missing Piece

- UDC's red and green phosphorescent emitters have been in mass production for years. Blue is the last unsolved piece an after years of R&D, LG Display has now verified UDC's blue PHOLED material works at mass-production scale.
- Every OLED panel using UDC's blue material means more material sold per panel and higher royalty income per device. Adoption triggers a global hardware upgrade cycle across billions of screens as device makers compete on battery life.
- Regulators, institutional investors, and device OEMs increasingly face pressure on power efficiency. Blue PHOLED delivers a compelling 25% further reduction in display power making it a regulatory and marketing imperative, not just an optional upgrade.
- As green finance mandates tighten globally, high-power legacy display technology becomes structurally disadvantaged. UDC sits at the intersection of energy efficiency and capital allocation which is a rare ESG-linked growth story in the semiconductors business.

Revenue Impact

Materials per Panel Increases	Adding blue PHOLED to every OLED panel means UDC sells three emitter colors instead of two leading a direct step-up in material revenue per unit shipped.
Royalty Rate Increases	New blue PHOLED licensing agreements will trigger a new layer of per-panel royalties on top of existing R+G royalty streams.
Upgrade Cycle Leads to UDC Material Volume Surge	Device makers will refresh OLED lines to adopt blue, driving short-term volume surge in UDC materials as the global installed base upgrades.

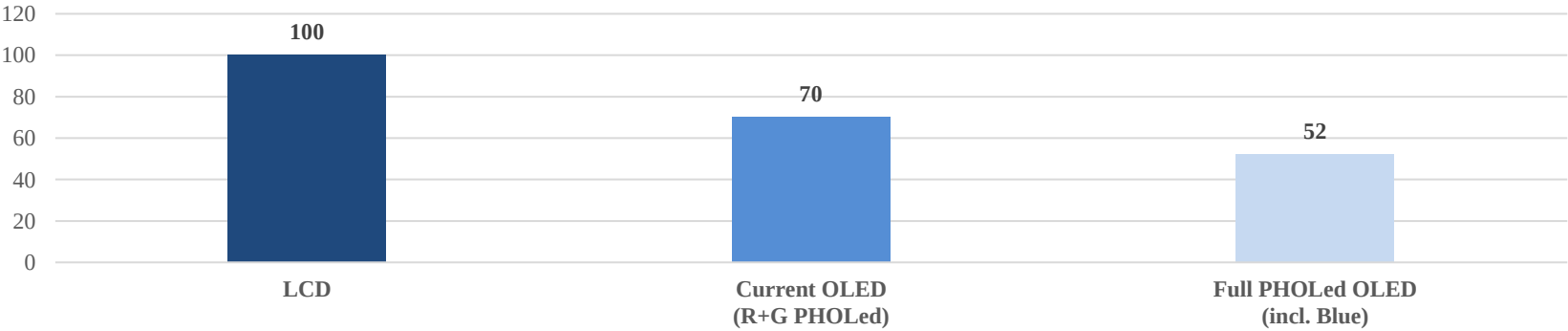
ESG & Power Efficiency

25% Power cut with per panel with Blue PHOLED

Aligns with green finance mandates

Billions of screens eligible for upgrade cycle

Relative Power Consumption



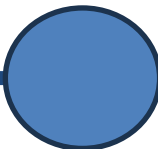
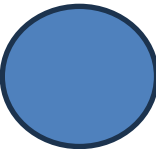
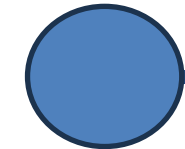
Path to Mass Adoption

2023-2024

2025

2026

2027+



Lab verification; LG Display confirms mass production viability

Panel maker qualification processes underway

First commercial devices with blue PHOLED expected

Adoption across smartphone & IT full royalty step-up



Risks & Catalysts

Catalysts/Drivers	Commentary
Gen 8.6 Fab Ramp (May 2026)	Samsung's A6 line begins mass production in May 2026, followed by BOE's B16 later this year. These fabs produce large-format IT panels; a single laptop panel requires 3x–5x more material than a smartphone, driving a massive volume surge.
Apple "Tandem" Expansion	Apple is standardising Tandem OLED (dual-stack layers) across the iPad and MacBook Pro lines. By stacking two layers of red, green, and blue material, UDC effectively doubles its revenue per device regardless of total unit sales growth.
Shareholder Returns & FCF	In Q1 2026, UDC increased its dividend by 11% to \$0.50/share. This, combined with active share buybacks, signals management's confidence that the heavy R&D cycle for Blue PHOLED is funded and the company is entering a high-cash-harvest phase.

Risk	Commentary
Blue PHOLED Technical Delays	Despite years of "preparations," actual commercial integration of Blue PHOLED remains elusive as of March 2026. If competitors like Lordin or Kyulux commercialise their alternatives first, UDC loses its "holy grail" monopoly on the full OLED stack.
2027 Patent Cliff Concerns	UDC's foundational phosphorescent patents begin expiring in 2027–2028. This creates a "generic threat" where Chinese panel makers could theoretically source cheaper materials from local suppliers, potentially compressing UDC's 76% gross margins.
Inventory & Guidance Misses	UDC's 2026 revenue guidance came in slightly below consensus due to "inventory digestion" from Chinese customers. High customer concentration means any delay in a single product launch causes significant stock volatility.



Valuation -DCF

Sensitivity Analysis

WACC	Perpetual Growth Rate					
	\$ 112.85	2.00%	2.25%	2.50%	2.75%	3.00%
	11.83%	\$ 94.80	\$ 96.21	\$ 97.69	\$ 99.25	\$ 100.90
	11.09%	\$ 101.12	\$ 102.81	\$ 104.60	\$ 106.50	\$ 108.52
	10.34%	\$ 108.57	\$ 110.64	\$ 112.85	\$ 115.19	\$ 117.70
	9.60%	\$ 117.50	\$ 120.08	\$ 122.83	\$ 125.79	\$ 128.97
	8.85%	\$ 128.39	\$ 131.65	\$ 135.17	\$ 138.98	\$ 143.11

Tax rate	Revenue Growth					
	\$ 112.85	-25.00%	-12.50%	0%	+12.50%	+25.00%
	24.12%	\$ 87.62	\$ 100.24	\$ 112.85	\$ 125.47	\$ 138.09
	22.56%	\$ 87.62	\$ 100.23	\$ 112.85	\$ 125.47	\$ 138.08
	21.00%	\$ 87.61	\$ 100.23	\$ 112.85	\$ 125.46	\$ 138.08
	19.44%	\$ 87.61	\$ 100.23	\$ 112.84	\$ 125.46	\$ 138.07
	17.88%	\$ 87.61	\$ 100.22	\$ 112.84	\$ 125.45	\$ 138.07

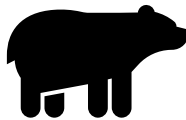
BULL – BASE – BEAR Scenarios



- The success of Blue PHOLED expansion and OLED integration surpasses expectations, increasing revenue from 2026 to 2030. The trailing 10-year average for the US Treasury Yield falls as inflation concerns fall, as Corporate Tax rates fall under a republican government, along with long-term growth rates increasing.



- Blue PHOLED success reaches expectations, along with OLED integration from 2026 to 2030. The trailing 10-year US Treasury Yield average remains, as corporate tax rates continue under a republican government, along with 5-year post-COVID growth rates within the US economy and OLED market remaining towards 2030.



- Blue PHOLED expansion underperforms expectations as buyer integration stagnates. US Treasury Yields rise as inflationary risks continue over wars in the Middle East, and corporate tax rates increase if a liberal government is elected, and the U.S. economy underperforms, along with the OLED market.

Scenario Analysis Summary

	BULL	BASE	BEAR
Revenue growth expectations	+25%	0%	-25%
Tax Rate	17.88%	21.00%	24.12%
Risk Free Rate	3.37%	3.62%	3.87%
Perpetual Growth rate	3.00%	2.50%	2.00%
Probability	25%	50%	25%
Return	57.66%	19.25%	-13.09%
Weighted Return	20.77%		



Valuation – Comparable Analysis

Company Name	Revenue Growth		EBITDA Margin		Profit Margin	
	23'-24'	24'-25'	24'	25'	24'	25'
OLED TECHNOLOGIES	12.36%	0.45%	43.68%	45.37%	34.29%	37.21%
InterDigital	58.03%	-3.97%	57.13%	64.56%	41.29%	48.75%
Dolby Laboratories	-1.92%	5.89%	28.89%	27.28%	20.57%	18.90%
Entegris	-8.03%	-1.36%	28.36%	28.75%	5.58%	9.16%
Veeco Instruments	7.64%	-7.39%	12.84%	10.76%	10.27%	5.33%
Mean	13.93%	-1.71%	31.80%	32.84%	19.43%	20.54%
Median	2.86%	-2.67%	28.62%	28.01%	15.42%	14.03%

Implied Value Per Share	Equity Values		
	25' REVENUE	25' EBITDA	25' EPS
Minimum	\$ 2,675,754,223.56	\$ 4,399,393,656.96	\$ 4,400,863,656.96
Mean	\$ 4,654,742,457.82	\$ 6,537,063,031.71	\$ 8,291,626,575.72
Median	\$ 4,232,709,767.98	\$ 6,455,145,206.83	\$ 5,437,754,457.84
Maximum	\$ 7,477,796,071.76	\$ 8,156,826,718.55	\$ 13,551,376,676.28

Trading Multiples					
EV/Revenue		EV/EBITDA		P/E	
24'	25'	24'	25'	24'	25'
5.29x	5.27x	12.12x	11.61x	19.82x	18.18x
9.60x	10.00x	16.81x	15.49x	25.45x	22.45x
4.02x	3.80x	13.92x	13.92x	21.86x	22.46x
6.14x	6.23x	21.67x	21.67x	90.62x	55.98x
2.43x	2.62x	18.90x	24.34x	25.07x	52.19x
5.55x	5.66x	17.82x	18.86x	40.75x	38.27x
5.08x	5.01x	17.85x	18.58x	25.26x	37.33x

Implied Value per share		
EV/Revenue	EV/EBITDA	P/E
\$56.81	\$93.41	\$93.44
\$98.83	\$138.80	\$176.05
\$89.87	\$137.06	\$115.46
\$158.77	\$173.19	\$287.73



Appendix - DCF

OLED DCF

Assumptions	
Revenue Growth	100.00%
Tax Rate	21.00%
Rf	3.62%
Perpetual Growth Rate (tp+)	2.50%
WACC	10.34%
Shares Outstanding	47,098,284
Current Share Price	\$93.44

(+/-)	1. BULL	2. BASE	3. BEAR
25.00%	125.00%	100.00%	75.00%
3.12%	17.88%	21.00%	24.12%
0.25%	3.37%	3.62%	3.87%
0.50%	3.00%	2.50%	2.00%

Choose Scenario -> 2

Year		FY2025	FY2026	FY2027	FY2028	FY2029	FY2030
Time Period	units	0	1	2	3	4	5
EBIT	\$	248,578,000.00	240,805,900.30	322,696,309.75	360,224,586.08	399,173,569.24	433,876,179.01
NOPAT	\$	196,376,620.00	190,236,661.24	254,930,084.70	284,577,423.01	315,347,119.70	342,762,181.42
(+) Depreciation and Amorization	\$	42,000,000.00	46,066,746.78	45,827,290.39	50,451,399.85	56,059,832.41	62,413,898.70
(-) CAPEX	\$	39,157,000.00	47,682,128.00	49,082,326.90	58,771,693.46	61,093,175.35	63,506,355.78
Change in Net Operating Working Capital	\$	82,874,000.00	-4,777,523.96	34,937,189.14	44,514,185.22	44,630,660.36	42,518,693.29
Unlevered FCF	\$	168,547,000.00	243,968,043.04	284,504,084.10	307,390,107.26	349,509,565.94	390,265,028.64
Terminal Value	\$						5,101,554,552.94
Discount Rate		1	0.91	0.82	0.74	0.67	0.61
Levered FCF	\$	168,547,000.00	221,103,364.39	233,675,576.08	228,811,112.03	235,780,943.09	238,600,680.75

Present Value of FCF (2026-30)	1,157,971,676.32
Present Value of Terminal Value	3,118,994,272.80
Enterprise Value (2026-2030)	4,276,965,949.13
(+) Cash	995,000,000.00
(-) Debt	23,970,000.00
Equity Value	5,247,995,949.13
(/) Shares Outstanding	47,098,284.00
Implied Value Per Share	\$ 111.43
Market Cap.	4,400,863,656.96
Intrinsic Value premium to market cap.	-847,132,292.17
	19.25%

DCF Components						
Revenue	676,105,892.20	772,794,831.26	843,410,914.34	916,628,355.89	986,527,459.42	
EBIT	240,805,900.30	322,696,309.75	360,224,586.08	399,173,569.24	433,876,179.01	
D&A	46,066,746.78	45,827,290.39	50,451,399.85	56,059,832.41	62,413,898.70	
CAPEX	47,682,128.00	49,082,326.90	58,771,693.46	61,093,175.35	63,506,355.78	
NOWC	-4,777,523.96	34,937,189.14	44,514,185.22	44,630,660.36	42,518,693.29	

Scenario Analysis		
BULL	BASE	BEAR
147.32	111.43	81.21
25%	50%	25%
	112.85	
	20.77%	

Weighted valuation	
DCF	COMPS
112.85	137.08
70%	30%
20.77%	46.70%
28.55%	
\$120.12	

Weighted Average Cost of Capital	
Market Value of Equity (E)	4,400,863,656,960.00
Market Value of Debt (D)	23,960,000,000.00
Total Market Value (V)	4,424,823,656,960.00
E/V	0.9946
D/V	0.0054
Cost of Equity (rE)	10.38%
Risk Free Rate (Rf)	3.62%
Beta (β)	1.23
Equity Risk Premium	5.50%
Cost of Debt (rD)	3.83%
WACC	10.34%

Perpetual Growth Rate						
	\$ 112.85	2.00%	2.25%	2.50%	2.75%	3.00%
WACC	11.83%	\$ 94.80	\$ 96.21	\$ 97.69	\$ 99.25	\$ 100.90
	11.09%	\$ 101.12	\$ 102.81	\$ 104.60	\$ 106.50	\$ 108.52
	10.34%	\$ 108.57	\$ 110.64	\$ 112.85	\$ 115.19	\$ 117.70
	9.60%	\$ 117.50	\$ 120.08	\$ 122.83	\$ 125.79	\$ 128.97
	8.85%	\$ 128.39	\$ 131.65	\$ 135.17	\$ 138.98	\$ 143.11

Revenue Growth						
	\$ 112.85	75.00%	87.50%	100.00%	112.50%	125.00%
Tax rate	24.12%	\$ 87.62	\$ 100.24	\$ 112.85	\$ 125.47	\$ 138.09
	22.56%	\$ 87.62	\$ 100.23	\$ 112.85	\$ 125.47	\$ 138.08
	21.00%	\$ 87.61	\$ 100.23	\$ 112.85	\$ 125.46	\$ 138.08
	19.44%	\$ 87.61	\$ 100.23	\$ 112.84	\$ 125.46	\$ 138.07
	17.88%	\$ 87.61	\$ 100.22	\$ 112.84	\$ 125.45	\$ 138.07



Appendix - IS

Income Statement											
Date		31/12/2021	31/12/2022	31/12/2023	31/12/2024	31/12/2025	2026E	2027E	2028E	2029E	2030E
Time in periods (thousands)	Units	0	1	2	3	4	5	6	7	8	9
Total Revenue	\$	553,525,000.00	616,619,000.00	576,429,000.00	647,684,000.00	650,611,000.00	676,105,892.20	772,794,831.26	843,410,914.34	916,628,355.89	986,527,459.42
Cost of Revenue	\$	114,991,000.00	127,896,000.00	135,367,000.00	148,461,000.00	154,126,000.00	172,407,002.51	172,487,806.34	189,598,773.54	206,058,054.40	221,771,372.88
Gross Profit	\$	438,534,000.00	488,723,000.00	441,062,000.00	499,223,000.00	496,485,000.00	503,698,889.69	600,307,024.92	653,812,140.80	710,570,301.48	764,756,086.54
Operating Expenses	\$	210,890,000.00	221,613,000.00	223,864,000.00	260,420,000.00	247,907,000.00	262,892,989.39	277,610,715.17	293,587,554.71	311,396,732.24	330,879,907.53
Research and Developement	\$	99,673,000.00	117,062,000.00	130,481,000.00	157,187,000.00	146,097,000.00	158,515,245.00	175,888,515.85	190,469,673.82	206,259,609.78	223,358,531.43
SG&A	\$	80,372,000.00	77,886,000.00	67,387,000.00	74,286,000.00	74,318,000.00	80,635,030.00	82,489,635.69	84,386,897.31	86,327,795.95	88,313,335.26
Amoriztaion of acquired tech and other intangibles	\$	21,994,000.00	17,459,000.00	15,993,000.00	18,200,000.00	18,198,000.00	13,300,000.00	8,500,000.00	7,626,000.00	7,319,000.00	7,319,000.00
Patent Costs	\$	8,160,000.00	8,329,000.00	9,356,000.00	8,699,000.00	8,790,000.00	9,574,738.39	9,906,981.81	10,250,754.08	10,606,455.25	10,974,499.24
Royalty and License expense	\$	691,000.00	877,000.00	647,000.00	2,048,000.00	504,000.00	867,976.00	825,581.82	854,229.51	883,871.27	914,541.60
Operating Profit	\$	227,644,000.00	267,110,000.00	217,198,000.00	238,803,000.00	248,578,000.00	240,805,900.30	322,696,309.75	360,224,586.08	399,173,569.24	433,876,179.01
Operating Income (EBIT)	\$	227,644,000.00	267,110,000.00	217,198,000.00	238,803,000.00	248,578,000.00	240,805,900.30	322,696,309.75	360,224,586.08	399,173,569.24	433,876,179.01
Interest Income, net	\$	505,000.00	7,811,000.00	28,166,000.00	40,682,000.00	39,708,000.00	40,621,284.00	41,555,573.53	42,511,351.72	43,489,112.81	44,489,362.41
Other Income (loss), net	\$	98,000.00	- 6,691,000.00	- 184,000.00	- 7,357,000.00	6,510,000.00	6,366,780.00	6,226,710.84	6,089,723.20	5,955,749.29	5,824,722.81
Interest and other Income, net	\$	603,000.00	1,120,000.00	27,982,000.00	33,325,000.00	46,218,000.00	46,988,064.00	47,782,284.37	48,601,074.92	49,444,862.10	50,314,085.21
Pretax Income	\$	228,247,000.00	268,230,000.00	245,180,000.00	272,128,000.00	294,796,000.00	334,782,028.30	418,260,878.50	457,426,735.93	498,063,293.45	534,504,349.44
Provision for income taxes	\$	44,034,000.00	58,169,000.00	42,160,000.00	50,049,000.00	52,721,000.00	63,608,585.38	79,469,566.91	86,911,079.83	94,632,025.76	101,555,826.39
Net Income	\$	184,213,000.00	210,061,000.00	203,020,000.00	222,079,000.00	242,075,000.00	271,173,442.92	338,791,311.58	370,515,656.11	403,431,267.69	432,948,523.05

Income Statement yoy growth %											
Date		31/12/2021	31/12/2022	31/12/2023	31/12/2024	31/12/2025	2026E	2027E	2028E	2029E	2030E
Time in periods (thousands)	Units	0	1	2	3	4	5	6	7	8	9
Total Revenue		553,525,000.00	616,619,000.00	576,429,000.00	647,684,000.00	650,611,000.00	676,105,892.20	772,794,831.26	843,410,914.34	916,628,355.89	986,527,459.42
year-on-year growth %			11.40%	-6.52%	12.36%	0.45%	3.92%	0.14	0.09	0.09	0.08
Cost of Revenue		114,991,000.00	127,896,000.00	135,367,000.00	148,461,000.00	154,126,000.00	172,407,002.51	172,487,806.34	189,598,773.54	206,058,054.40	221,771,372.88
year-on-year growth %			11.22%	5.84%	9.67%	3.82%	11.86%	0.00	0.10	0.09	0.08
Gross Profit		438,534,000.00	488,723,000.00	441,062,000.00	499,223,000.00	496,485,000.00	503,698,889.69	600,307,024.92	653,812,140.80	710,570,301.48	764,756,086.54
year-on-year growth %			11.44%	-9.75%	13.19%	-1.45%	1.45%	0.19	0.09	0.09	0.08
Operating Expenses		210,890,000.00	221,613,000.00	223,864,000.00	260,420,000.00	247,907,000.00	262,892,989.39	277,610,715.17	293,587,554.71	311,396,732.24	330,879,907.53
year-on-year growth %			5.08%	1.02%	16.33%	-4.80%	6.05%	0.06	0.06	0.06	0.06
Research and Developement		99,673,000.00	117,062,000.00	130,481,000.00	157,187,000.00	146,097,000.00	158,515,245.00	175,888,515.85	190,469,673.82	206,259,609.78	223,358,531.43
year-on-year growth %			17.45%	11.46%	20.47%	-7.06%	8.50%	0.11	0.08	0.08	0.08
SG&A		80,372,000.00	77,886,000.00	67,387,000.00	74,286,000.00	74,318,000.00	80,635,030.00	82,489,635.69	84,386,897.31	86,327,795.95	88,313,335.26
year-on-year growth %			-3.09%	-13.48%	10.24%	0.04%	8.50%	0.02	0.02	0.02	0.02
Amoriztaion of acquired tech and other intangibles		21,994,000.00	17,459,000.00	15,993,000.00	18,200,000.00	18,198,000.00	13,300,000.00	8,500,000.00	7,626,000.00	7,319,000.00	7,319,000.00
year-on-year growth %			-20.62%	-8.40%	13.80%	-0.01%	-26.92%	-0.36	-0.10	-0.04	0.00
Patent Costs		8,160,000.00	8,329,000.00	9,356,000.00	8,699,000.00	8,790,000.00	9,574,738.39	9,906,981.81	10,250,754.08	10,606,455.25	10,974,499.24
year-on-year growth %			2.07%	12.33%	-7.02%	1.05%	8.93%	0.03	0.03	0.03	0.03
Royalty and License expense		691,000.00	877,000.00	647,000.00	2,048,000.00	504,000.00	867,976.00	825,581.82	854,229.51	883,871.27	914,541.60
year-on-year growth %			26.92%	-26.23%	216.54%	-75.39%	72.22%	-0.05	0.03	0.03	0.03
Operating Profit		227,644,000.00	267,110,000.00	217,198,000.00	238,803,000.00	248,578,000.00	240,805,900.30	322,696,309.75	360,224,586.08	399,173,569.24	433,876,179.01
year-on-year growth %			17.34%	-18.69%	9.95%	4.09%	-3.13%	0.34	0.12	0.11	0.09
Operating Income (EBIT)		227,644,000.00	267,110,000.00	217,198,000.00	238,803,000.00	248,578,000.00	240,805,900.30	322,696,309.75	360,224,586.08	399,173,569.24	433,876,179.01
year-on-year growth %			17.34%	-18.69%	9.95%	4.09%	-3.13%	0.34	0.12	0.11	0.09
Interest Income, net		505,000.00	7,811,000.00	28,166,000.00	40,682,000.00	39,708,000.00	40,621,284.00	41,555,573.53	42,511,351.72	43,489,112.81	44,489,362.41
year-on-year growth %			1446.73%	260.59%	44.44%	-2.39%	2.30%	0.02	0.02	0.02	0.02
Other Income (loss), net		98,000.00	-6,691,000.00	-184,000.00	-7,357,000.00	6,510,000.00	6,366,780.00	6,226,710.84	6,089,723.20	5,955,749.29	5,824,722.81
year-on-year growth %			-6927.55%	-97.25%	3898.37%	-188.49%	-2.20%	-0.02	-0.02	-0.02	-0.02
Interest and other Income, net		603,000.00	1,120,000.00	27,982,000.00	33,325,000.00	46,218,000.00	46,988,064.00	47,782,284.37	48,601,074.92	49,444,862.10	50,314,085.21
year-on-year growth %			85.74%	2398.39%	19.09%	38.69%	1.67%	0.02	0.02	0.02	0.02
Pretax Income		228,247,000.00	268,230,000.00	245,180,000.00	272,128,000.00	294,796,000.00	334,782,028.30	418,260,878.50	457,426,735.93	498,063,293.45	534,504,349.44
year-on-year growth %			17.52%	-8.59%	10.99%	8.33%	13.56%	0.25	0.09	0.09	0.07
Provision for income taxes		44,034,000.00	58,169,000.00	42,160,000.00	50,049,000.00	52,721,000.00	63,608,585.38	79,469,566.91	86,911,079.83	94,632,025.76	101,555,826.39
year-on-year growth %			32.10%	-27.52%	18.71%	5.34%	20.65%	0.25	0.09	0.09	0.07
Net Income		184,213,000.00	210,061,000.00	203,020,000.00	222,079,000.00	242,075,000.00	271,173,442.92	338,791,311.58	370,515,656.11	403,431,267.69	432,948,523.05
year-on-year growth %			14.03%	-3.35%	9.39%	9.00%	12.02%	0.25	0.09	0.09	0.07



Appendix - BS

Balance Sheet											
Date		31/12/2021	31/12/2022	31/12/2023	31/12/2024	31/12/2025	2026E	2027E	2028E	2029E	2030E
Time in periods (thousands)	Units	0	1	2	3	4	5	6	7	8	9
Current Assets											
Cash and Cash Equivalents	\$	311,993,000.00	93,430,000.00	91,985,000.00	98,980,000.00	138,353,000.00	390,431,715.29	607,115,286.81	809,414,956.80	1,014,402,154.83	1,217,585,513.22
Short-term investments	\$	351,194,000.00	484,345,000.00	422,137,000.00	393,690,000.00	464,004,000.00	355,630,121.24	392,184,844.15	394,627,458.54	408,485,918.50	432,871,256.04
Accounts receivables	\$	107,639,000.00	92,664,000.00	139,850,000.00	113,648,000.00	119,953,000.00	128,080,284.81	146,396,863.61	159,774,247.45	173,644,427.96	186,885,988.44
Inventory	\$	134,160,000.00	183,220,000.00	175,795,000.00	182,938,000.00	240,912,000.00	230,792,257.12	230,900,424.99	253,805,983.84	275,839,164.19	296,873,763.63
Other Current Assets	\$	20,948,000.00	45,791,000.00	87,365,000.00	110,575,000.00	123,836,000.00	125,079,590.06	140,648,659.29	153,500,786.41	166,826,360.77	179,547,997.61
Total Current Assets	\$	925,934,000.00	899,450,000.00	917,132,000.00	899,831,000.00	1,087,058,000.00	1,230,013,968.52	1,517,246,078.86	1,771,123,433.05	2,039,198,026.24	2,313,764,518.98
Property and Equipment (net)	\$	128,832,000.00	143,445,000.00	175,150,000.00	195,239,000.00	214,947,000.00	229,862,381.22	241,617,417.73	257,563,711.34	269,916,054.28	278,327,511.36
Acquired Technology (net)	\$	49,668,000.00	38,382,000.00	90,325,000.00	73,554,000.00	56,783,000.00	-14,679,000.00	-123,580,000.00	-130,780,000.00	-137,880,000.00	-144,980,000.00
Othe intangible assets (net)	\$	9,711,000.00	8,247,000.00	6,874,000.00	5,446,000.00	4,019,000.00	2,975,000.00	-244,000.00	-5,900,000.00	-11,348,000.00	-11,535,000.00
Goodwill	\$	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00	15,535,000.00
Investments	\$	168,076,000.00	259,861,000.00	299,548,000.00	457,593,000.00	377,034,000.00	476,911,282.38	525,932,326.27	529,207,949.71	547,792,584.46	580,494,096.24
Deffered Income Taxes	\$	33,453,000.00	58,161,000.00	59,108,000.00	78,320,000.00	79,454,000.00	86,979,925.05	91,428,030.87	97,462,108.37	102,136,234.94	105,319,130.30
Other Assets	\$	135,710,000.00	109,739,000.00	105,289,000.00	106,815,000.00	128,932,000.00	122,307,555.90	139,798,584.98	152,573,034.40	165,818,069.58	178,462,817.41
Total Assets	\$	1,466,919,000.00	1,532,820,000.00	1,668,961,000.00	1,832,333,000.00	1,963,762,000.00	2,149,906,113.07	2,407,733,438.70	2,686,785,236.87	2,991,167,969.50	3,315,388,074.28
Current liabilities											
Accounts payable	\$	14,955,000.00	9,519,000.00	10,933,000.00	36,590,000.00	23,344,000.00	23,556,639.96	23,567,680.50	25,905,618.57	28,154,514.19	30,301,486.07
Accured expenses	\$	45,474,000.00	51,002,000.00	52,080,000.00	46,026,000.00	52,564,000.00	57,839,507.02	55,507,275.86	56,783,943.20	58,089,973.89	59,426,043.29
Deferred revenue	\$	120,864,000.00	45,599,000.00	47,713,000.00	33,074,000.00	21,011,000.00	20,283,176.77	23,183,844.94	25,302,327.43	27,498,850.68	29,595,823.78
Other current liabilities	\$	6,645,000.00	29,577,000.00	8,096,000.00	9,720,000.00	11,094,000.00	9,634,508.96	11,012,326.35	12,018,605.53	13,061,954.07	14,058,016.30
Total current liabilities	\$	187,938,000.00	135,697,000.00	118,822,000.00	125,410,000.00	108,013,000.00	111,313,832.71	113,271,127.64	120,010,494.73	126,805,292.83	133,381,369.44
Deferred Revenue	\$	36,217,000.00	18,279,000.00	12,006,000.00	537,000.00	1,943,000.00	1,238,103.00	4,874,963.83	4,221,361.64	4,985,763.74	5,914,729.73
Retirement plan benefit liability	\$	66,773,000.00	59,790,000.00	52,249,000.00	54,450,000.00	56,541,000.00	58,241,000.00	60,141,000.00	62,241,000.00	64,541,000.00	66,841,000.00
Other liabilities	\$	76,077,000.00	43,685,000.00	38,658,000.00	35,411,000.00	36,246,000.00	41,986,175.91	47,990,559.02	52,375,817.78	56,922,620.90	61,263,355.23
Total liabilities	\$	367,005,000.00	257,451,000.00	221,735,000.00	215,808,000.00	202,743,000.00	212,779,111.62	226,277,650.49	238,848,674.15	253,254,677.47	267,400,454.40
Shareholders Equity											
Preffered Stock	\$	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00	2,000.00
Common Stock	\$	491,000.00	491,000.00	487,000.00	488,000.00	489,000.00	490,000.00	491,000.00	492,000.00	493,000.00	494,000.00
Additional paid in capital	\$	658,728,000.00	681,335,000.00	699,554,000.00	723,719,000.00	744,692,000.00	753,692,000.00	762,692,000.00	771,692,000.00	780,692,000.00	789,692,000.00
retained earnings	\$	500,212,000.00	653,277,000.00	789,553,000.00	934,655,000.00	1,090,479,000.00	1,258,036,218.12	1,493,211,304.90	1,750,691,079.41	2,031,666,808.70	2,332,740,136.55
Accumulated other comprehensive income (loss	\$ -	18,235,000.00 -	18,452,000.00 -	1,086,000.00 -	1,055,000.00	781,000.00	453,333.33	453,333.33	453,333.33	453,333.33	453,333.33
Treasury stock	\$ -	41,284,000.00 -	41,284,000.00 -	41,284,000.00 -	41,284,000.00 -	75,424,000.00	-75,424,000.00	-75,424,000.00	-75,424,000.00	-75,424,000.00	-75,424,000.00
Total Shareholder Equity	\$	1,099,914,000.00	1,275,369,000.00	1,447,226,000.00	1,616,525,000.00	1,761,019,000.00	1,937,249,551.45	2,181,425,638.23	2,447,906,412.74	2,737,883,142.03	3,047,957,469.88
Total Shareholders equity and liabilities	\$	1,466,919,000.00	1,532,820,000.00	1,668,961,000.00	1,832,333,000.00	1,963,762,000.00	2,150,028,663.07	2,407,703,288.73	2,686,755,086.89	2,991,137,819.51	3,315,357,924.28



Appendix - CF

Cash Flow Statement											
Date		31/12/2021	31/12/2022	31/12/2023	31/12/2024	31/12/2025	2026E	2027E	2028E	2029E	2030E
Time in periods (thousands)	Units	0	1	2	3	4	5	6	7	8	9
Cash Flows from Operating activities:											
Net Income	\$	184,213,000.00	210,061,000.00	203,020,000.00	222,079,000.00	242,075,000.00	271,173,442.92	338,791,311.58	370,515,656.11	403,431,267.69	432,948,523.05
Adjustments:	\$	88,860,000.00	50,550,000.00	66,874,000.00	56,962,000.00	68,284,000.00	60,848,686.88	66,921,441.98	71,939,524.49	80,912,578.76	90,508,908.92
Depreciation	\$	19,968,000.00	24,815,000.00	27,409,000.00	25,940,000.00	28,389,000.00	32,766,746.78	37,327,290.39	42,825,399.85	48,740,832.41	55,094,898.70
Impairment of property and equipment	\$	-	-	-	7,498,000.00	1,577,000.00	0.00	0.00	0.00	0.00	0.00
Amortization of intangibles	\$	21,994,000.00	17,459,000.00	15,993,000.00	18,200,000.00	18,198,000.00	13,300,000.00	8,500,000.00	7,626,000.00	7,319,000.00	7,319,000.00
Investment gains, net	\$	-	-	-	7,399,000.00	8,765,000.00	-9,641,500.00	-10,605,650.00	-11,666,215.00	-12,832,836.50	-14,116,120.15
Stock-based compensation	\$	36,275,000.00	29,946,000.00	24,109,000.00	30,032,000.00	28,227,000.00	30,424,765.15	34,775,767.41	37,953,491.15	41,248,276.01	44,393,735.67
Deferred income tax benefit	\$	1,748,000.00	26,946,000.00	3,766,000.00	19,117,000.00	1,036,000.00	-7,525,925.05	-4,448,105.82	-6,034,077.50	-4,674,126.57	-3,182,895.36
Retirement plan expense, net of benefit payments	\$	8,875,000.00	5,276,000.00	3,129,000.00	1,808,000.00	1,694,000.00	1,524,600.00	1,372,140.00	1,234,926.00	1,111,433.40	1,000,290.06
Decrease (increase) in assets:	\$	- 100,518,000.00	- 32,957,000.00	- 76,885,000.00	- 7,119,000.00	- 91,234,000.00	7,373,312.11	-51,484,844.98	-61,909,519.24	-62,473,970.39	-59,642,544.63
Accounts receivable	\$	- 25,378,000.00	14,975,000.00	- 47,186,000.00	26,202,000.00	- 6,305,000.00	-8,127,284.81	-18,316,578.80	-13,377,383.84	-13,870,180.51	-13,241,560.52
Inventory	\$	- 42,569,000.00	- 49,060,000.00	7,425,000.00	- 7,143,000.00	- 57,974,000.00	10,119,742.88	-108,167.87	-22,905,558.86	-22,033,180.35	-21,034,599.43
Other current assets	\$	- 202,000.00	- 24,843,000.00	- 41,574,000.00	- 23,210,000.00	- 3,261,000.00	-1,243,590.06	-15,569,069.23	-12,852,127.12	-13,325,574.36	-12,721,636.84
Other assets	\$	- 32,369,000.00	25,971,000.00	4,450,000.00	- 2,968,000.00	- 23,694,000.00	6,624,444.10	-17,491,029.08	-12,774,449.43	-13,245,035.18	-12,644,747.83
Increase (decrease) in liabilities:	\$	18,923,000.00	- 101,340,000.00	- 26,620,000.00	- 18,182,000.00	- 8,295,000.00	-225,258.12	-3,165,862.12	-312,052.23	925,700.00	1,172,183.82
Accounts payable and accrued expenses	\$	1,902,000.00	3,338,000.00	4,047,000.00	10,357,000.00	2,114,000.00	5,488,146.98	-2,321,190.62	3,614,605.41	3,554,926.32	3,483,041.28
Other current liabilities	\$	2,105,000.00	20,917,000.00	- 21,481,000.00	1,269,000.00	- 441,000.00	1,459,491.04	-1,377,817.38	-1,006,279.18	-1,043,348.54	-996,062.23
Deferred revenue	\$	- 5,220,000.00	- 93,203,000.00	- 4,159,000.00	- 26,108,000.00	- 10,657,000.00	-1,432,720.23	6,537,529.00	1,464,880.30	2,960,925.35	3,025,939.09
Other liabilities	\$	20,136,000.00	- 32,392,000.00	- 5,027,000.00	- 3,700,000.00	689,000.00	-5,740,175.91	-6,004,383.12	-4,385,258.76	-4,546,803.12	-4,340,734.33
Net cash provided by operating activities	\$	191,478,000.00	126,314,000.00	166,389,000.00	253,740,000.00	210,830,000.00	339,170,183.79	351,062,046.46	380,233,609.13	422,795,576.06	464,987,071.16
Cash Flows from investing activities:											
Purchases of property and equipment	\$	- 43,161,000.00	- 42,497,000.00	- 59,792,000.00	- 42,637,000.00	- 56,470,000.00	-47,682,128.00	-49,082,326.90	-58,771,693.46	-61,093,175.35	-63,506,355.78
Purchase of intangibles	\$	- 394,000.00	- 4,709,000.00	- 66,563,000.00	-	- 10,000,000.00	-	-	-	-	-
Purchases of investments	\$	- 642,180,000.00	- 701,993,000.00	- 531,103,000.00	- 594,848,000.00	- 375,506,000.00	-413,056,600.00	-495,667,920.00	-545,234,712.00	-599,758,183.20	-659,734,001.52
Proceeds from sale and maturity of investments	\$	227,984,000.00	468,456,000.00	574,165,000.00	473,075,000.00	396,500,000.00	475,800,000.00	523,380,000.00	549,549,000.00	577,026,450.00	605,877,772.50
Net cash used in investing activities	\$	- 457,751,000.00	- 280,743,000.00	- 83,293,000.00	- 164,410,000.00	- 45,476,000.00	15,061,272.00	-21,370,246.90	-54,457,405.46	-83,824,908.55	-117,362,584.80
Cash Flow from financing activities:											
Proceeds from issuance of common stock	\$	1,507,000.00	1,570,000.00	2,012,000.00	2,220,000.00	2,020,000.00	2,084,000.00	2,084,000.00	2,084,000.00	2,084,000.00	2,084,000.00
Repurchases of common stock	\$	-	-	-	-	- 32,881,000.00	-	-	-	-	-
Payment of withholding taxes related to stock-based compensation to employees	\$	- 14,949,000.00	- 9,209,000.00	- 8,206,000.00	- 8,386,000.00	- 9,571,000.00	-10,040,172.50	-11,476,003.24	-12,524,652.08	-13,611,931.08	-14,649,932.77
Cash dividends paid	\$	- 37,931,000.00	- 56,996,000.00	- 66,735,000.00	- 76,169,000.00	- 85,549,000.00	-94,196,568.00	-103,616,224.80	-113,035,881.60	-122,455,538.40	-131,875,195.20
Net cash used in financing activities	\$	- 51,373,000.00	- 64,635,000.00	- 72,929,000.00	- 82,335,000.00	- 125,981,000.00	-102,152,740.50	-113,008,228.04	-123,476,533.68	-133,983,469.48	-144,441,127.97



Appendix – WC Schedule

Working Capital Schedule											
Date		31/12/2021	31/12/2022	31/12/2023	31/12/2024	31/12/2025	2026E	2027E	2028E	2029E	2030E
Time in periods (thousands)	Units	0	1	2	3	4	5	6	7	8	9
CURRENT ASSETS											
Cash and cash equivalents	\$	311,993,000.00	93,430,000.00	91,985,000.00	98,980,000.00	138,353,000.00	390,431,715.29	607,115,286.81	809,414,956.80	1,014,402,154.83	1,217,585,513.22
Short-term investments	\$	351,194,000.00	484,345,000.00	422,137,000.00	393,690,000.00	464,004,000.00	355630121.2	392184844.2	394627458.5	408485918.5	432871256
Accounts receivable, net	\$	107,639,000.00	92,664,000.00	139,850,000.00	113,648,000.00	119,953,000.00	128,080,284.81	146,396,863.61	159,774,247.45	173,644,427.96	186,885,988.48
Inventory	\$	134,160,000.00	183,220,000.00	175,795,000.00	182,938,000.00	240,912,000.00	230,792,257.12	230,900,424.99	253,805,983.84	275,839,164.19	296,873,763.63
Other current assets	\$	20,948,000.00	45,791,000.00	87,365,000.00	110,575,000.00	123,836,000.00	125,079,590.06	140,648,659.29	153,500,786.41	166,826,360.77	179,547,997.61
Total Current Assets	\$	925,934,000.00	899,450,000.00	917,132,000.00	899,831,000.00	1,087,058,000.00	1,230,013,968.52	1,517,246,078.86	1,771,123,433.05	2,039,198,026.24	2,313,764,518.98
CURRENT LIABILITIES											
Accounts payable	\$	14,955,000.00	9,519,000.00	10,933,000.00	36,590,000.00	23,344,000.00	23,556,639.96	23,567,680.50	25,905,618.57	28,154,514.19	30,301,486.07
Accrued expenses	\$	45,474,000.00	51,002,000.00	52,080,000.00	46,026,000.00	52,564,000.00	57,839,507.02	55,507,275.86	56,783,943.20	58,089,973.89	59,426,043.29
Deferred revenue (current)	\$	120,864,000.00	45,599,000.00	47,713,000.00	33,074,000.00	21,011,000.00	20,283,176.77	23,183,844.94	25,302,327.43	27,498,850.68	29,595,823.78
Other current liabilities	\$	6,645,000.00	29,577,000.00	8,096,000.00	9,720,000.00	11,094,000.00	9,634,508.96	11,012,326.35	12,018,605.53	13,061,954.07	14,058,016.30
Total Current Liabilities	\$	187,938,000.00	135,697,000.00	118,822,000.00	125,410,000.00	108,013,000.00	111,313,832.71	113,271,127.64	120,010,494.73	126,805,292.83	133,381,369.44
NET WORKING CAPITAL											
Net Working Capital (NWC)	\$	737,996,000.00	763,753,000.00	798,310,000.00	774,421,000.00	979,045,000.00	1,118,700,135.81	1,403,974,951.21	1,651,112,938.32	1,912,392,733.41	2,180,383,149.53
YoY Change in NWC			3.49%	4.52%	-2.99%	26.42%	14.26%	25.50%	17.60%	15.82%	14.01%
OPERATIONAL WORKING CAPITAL											
Operational NWC	\$	195,673,000.00	231,577,000.00	331,901,000.00	314,825,000.00	397,699,000.00	392,921,476.04	427,858,665.18	472,372,850.40	517,003,510.76	559,522,204.06
Change in NWC			35,904,000.00	100,324,000.00	- 17,076,000.00	82,874,000.00	- 4,777,523.96	34,937,189.14	44,514,185.22	44,630,660.36	42,518,693.29
EFFICIENCY / TURNOVER METRICS											
Revenue		553,525,000.00	616,619,000.00	576,429,000.00	647,684,000.00	650,611,000.00	676,105,892.20	772,794,831.26	843,410,914.34	916,628,355.89	986,527,459.42
Cost of Goods Sold		114,991,000.00	127,896,000.00	135,367,000.00	148,461,000.00	154,126,000.00	172,407,002.51	172,487,806.34	189,598,773.54	206,058,054.40	221,771,372.88
Days Sales Outstanding (DSO)		70.98	54.85	88.55	64.05	67.29	69.14	69.14	69.14	69.14	69.14
5-year Historical average						69.14					
Days Inventory Outstanding (DIO)		425.85	522.89	474.01	449.76	570.53	488.61	488.61	488.61	488.61	488.61
5-year Historical average						488.61					
Days Payable Outstanding (DPO)		47.47	27.17	29.48	89.96	55.28	49.87	49.87	49.87	49.87	49.87
5-year Historical average						49.87					
Cash Conversion Cycle (CCC)		449.35	550.57	533.08	423.85	582.54	507.88	507.88	507.88	507.88	507.88
NWC as % of Revenue											
		1.33	1.24	1.38	1.20	1.50	1.654622669	1.816749924	1.957661337	2.086333814	2.210159615
Current Ratio		4.93	6.63	7.72	7.18	10.06	11.04996512	13.39481747	14.75807126	16.0813321	17.34698428
Quick Ratio		4.21	5.28	6.24	5.72	7.83	8.976617614	11.35634191	12.64320635	13.90603517	15.12123293
OCA as a % of revenue		3.78%	7.43%	15.16%	17.07%	19.03%	18.50%	18.20%	18.20%	18.20%	18.20%
Deferred revenue as a % of revenue		21.84%	7.40%	8.28%	5.11%	3.23%	3.00%	3.00%	3.00%	3.00%	3.00%
Accrued expenses as a % of SG&A		56.58%	65.48%	77.28%	61.96%	70.73%	71.73%	67.29%	67.29%	67.29%	67.29%
OCL as a % of revenue		1.20%	4.80%	1.40%	1.50%	1.71%	1.43%	1.43%	1.43%	1.43%	1.43%



Appendix – D&A

Depreciation and Amortization Schedule											
Date		31/12/2021	31/12/2022	31/12/2023	31/12/2024	31/12/2025	2026E	2027E	2028E	2029E	2030E
Time in periods	units	0	1	2	3	4	5	6	7	8	9
PP&E — PROPERTY, PLANT & EQUIPMENT											
Gross PP&E — Beginning of Period	\$	174,606,000.00	221,293,000.00	260,563,000.00	319,058,000.00	365,116,000.00	404,273,000.00	451,955,128.00	501,037,454.90	559,809,148.36	620,902,323.71
(+) Capital Expenditures (CapEx)	\$	46,687,000.00	39,270,000.00	58,495,000.00	46,058,000.00	39,157,000.00	47,682,128.00	49,082,326.90	58,771,693.46	61,093,175.35	63,506,355.78
Gross PP&E — End of Period	\$	221,293,000.00	260,563,000.00	319,058,000.00	365,116,000.00	404,273,000.00	451,955,128.00	501,037,454.90	559,809,148.36	620,902,323.71	684,408,679.49
yoy growth %		26.74%	17.75%	22.45%	14.44%	10.72%	11.79%	10.86%	11.73%	12.80%	12.89%
Accumulated Depreciation — Beginning of Period	\$	72,493,000.00	92,461,000.00	117,118,000.00	143,908,000.00	169,877,000.00	189,326,000.00	222,092,746.78	259,420,037.17	302,245,437.02	350,986,269.43
(+) Depreciation Expense (current year)	\$	20,000,000.00	24,800,000.00	27,400,000.00	25,900,000.00	28,400,000.00	32,766,746.78	37,327,290.39	42,825,399.85	48,740,832.41	55,094,898.70
(-) Accumulated Depr. on Disposals	\$	32,000.00	143,000.00	610,000.00	69,000.00	8,951,000.00	-	-	-	-	-
Accumulated Depreciation — End of Period	\$	92,461,000.00	117,118,000.00	143,908,000.00	169,877,000.00	189,326,000.00	222,092,746.78	259,420,037.17	302,245,437.02	350,986,269.43	406,081,168.13
Net PP&E (Gross PP&E – Accumulated Depreciation)	\$	128,832,000.00	143,445,000.00	175,150,000.00	195,239,000.00	214,947,000.00	229,862,381.22	241,617,417.73	257,563,711.34	269,916,054.28	278,327,511.36
Weighted Average Useful Life Assumption (years)											
Implied Depreciation Rate (Dep. Expense / Avg Gross PP&E)		0.06	0.08	0.09	0.08	0.09	0.10	0.12	0.14	0.16	0.18
INTANGIBLE ASSETS — AMORTIZATION											
Gross Intangibles — Beginning of Period		16,840,000.00	17,276,000.00	17,236,000.00	17,288,000.00	17,288,000.00	17,288,000.00	15,340,000.00	10,080,000.00	4,820,000.00	4,820,000.00
Customer relationships	\$	10,520,000.00	10,520,000.00	10,520,000.00	10,520,000.00	10,520,000.00	10,520,000.00	10,520,000.00	5,260,000.00	-	-
Developed IP, processes and recepies	\$	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00	4,820,000.00
Trade name/Trademarks	\$	1,500,000.00	1,500,000.00	1,500,000.00	1,500,000.00	1,500,000.00	1,500,000.00	-	-	-	-
Other	\$	-	436,000.00	396,000.00	448,000.00	448,000.00	448,000.00	-	-	-	-
(+) Additions (acquisitions / patent purchases)	\$	436,000.00	-	52,000.00	-	-	-	-	-	-	-
(-) Disposals / Write-offs	\$	-	40,000.00	-	-	-	-	-	-	-	-
Gross Intangibles — End of Period	\$	17,276,000.00	17,316,000.00	17,288,000.00	17,288,000.00	17,288,000.00	17,288,000.00	15,340,000.00	10,080,000.00	4,820,000.00	4,820,000.00
Accumulated Amortization — Beginning of Period	\$	6,155,000.00	7,545,000.00	8,921,000.00	10,296,000.00	11,668,000.00	13,041,000.00	14,313,000.00	15,584,000.00	15,980,000.00	16,168,000.00
(+) Amortisation Expense (current year)	\$	1,400,000.00	1,400,000.00	1,400,000.00	1,400,000.00	1,400,000.00	1,300,000.00	1,300,000.00	426,000.00	219,000.00	219,000.00
(-) Accumulated Amort. on Disposals	\$	10,000.00	24,000.00	25,000.00	28,000.00	27,000.00	28,000.00	29,000.00	30,000.00	31,000.00	32,000.00
Accumulated Amortization — End of Period	\$	7,545,000.00	8,921,000.00	10,296,000.00	11,668,000.00	13,041,000.00	14,313,000.00	15,584,000.00	15,980,000.00	16,168,000.00	16,355,000.00
Net Intangibles (Gross – Accumulated Amortization)	\$	9,731,000.00	8,395,000.00	6,992,000.00	5,620,000.00	4,247,000.00	2,975,000.00	244,000.00	5,900,000.00	11,348,000.00	11,535,000.00
ACQUIRED TECHNOLOGY — AMORTIZATION											
Gross Goodwill — Beginning of Period	\$	223,303,000.00	223,303,000.00	228,053,000.00	277,175,000.00	277,175,000.00	327,175,000.00	212,001,000.00	116,012,000.00	116,012,000.00	116,012,000.00
PD-LD	\$	1,481,000.00	1,481,000.00	1,481,000.00	-	-	-	-	-	-	-
Motorola	\$	15,909,000.00	15,909,000.00	15,909,000.00	-	-	-	-	-	-	-
Merck KGaA	\$	-	-	-	66,012,000.00	66,012,000.00	116,012,000.00	116,012,000.00	116,012,000.00	116,012,000.00	116,012,000.00
BASF	\$	95,989,000.00	95,989,000.00	95,989,000.00	95,989,000.00	95,989,000.00	95,989,000.00	95,989,000.00	-	-	-
Fujifilm	\$	109,462,000.00	109,462,000.00	109,462,000.00	109,462,000.00	109,462,000.00	109,462,000.00	-	-	-	-
Other	\$	462,000.00	462,000.00	5,212,000.00	5,712,000.00	5,712,000.00	5,712,000.00	-	-	-	-
(+) Additions (acquisitions / patent purchases)	\$	-	-	66,512,000.00	-	50,000,000.00	-	-	-	-	-
(-) End of acquisition	\$	-	-	17,390,000.00	-	-	109,462,000.00	95,989,000.00	-	-	-
Gross Goodwill — End of Period	\$	223,303,000.00	223,303,000.00	277,175,000.00	277,175,000.00	327,175,000.00	217,713,000.00	116,012,000.00	116,012,000.00	116,012,000.00	116,012,000.00
Accumulated Amortization — Beginning of Period	\$	153,050,000.00	173,635,000.00	189,671,000.00	186,850,000.00	203,621,000.00	220,392,000.00	232,392,000.00	239,592,000.00	246,792,000.00	253,892,000.00
(+) Amortisation Expense (current year)	\$	20,600,000.00	16,000,000.00	14,600,000.00	16,800,000.00	16,800,000.00	12,000,000.00	7,200,000.00	7,200,000.00	7,100,000.00	7,100,000.00
(-) Accumulated Amort. on Disposals	\$	15,000.00	36,000.00	17,421,000.00	29,000.00	29,000.00	-	-	-	-	-
Accumulated Amortization — End of Period	\$	173,635,000.00	189,671,000.00	186,850,000.00	203,621,000.00	220,392,000.00	232,392,000.00	239,592,000.00	246,792,000.00	253,892,000.00	260,992,000.00
Net Acquired Technology (Gross – Accumulated Amortization)	\$	49,668,000.00	33,632,000.00	90,325,000.00	73,554,000.00	106,783,000.00	14,679,000.00	123,580,000.00	130,780,000.00	137,880,000.00	144,980,000.00
Implied Amortization Rate (Amort. Expense / Avg Gross Intangibles)		8.10%	8.10%	8.10%	8.10%	8.10%	7.52%	7.52%	2.46%	1.27%	1.27%
TOTAL D&A SUMMARY											
Depreciation of PP&E	\$	20,000,000.00	24,800,000.00	27,400,000.00	25,900,000.00	28,400,000.00	32,766,746.78	37,327,290.39	42,825,399.85	48,740,832.41	55,094,898.70
Amortisation of Intangibles	\$	1,400,000.00	1,400,000.00	1,400,000.00	1,400,000.00	1,400,000.00	1,300,000.00	1,300,000.00	426,000.00	219,000.00	219,000.00
Other Amortisation (Acquired Technology)	\$	20,600,000.00	16,000,000.00	14,600,000.00	16,800,000.00	16,800,000.00	12,000,000.00	7,200,000.00	7,200,000.00	7,100,000.00	7,100,000.00
Total D&A	\$	42,000,000.00	42,200,000.00	43,400,000.00	44,100,000.00	46,600,000.00	46,066,746.78	45,827,290.39	50,451,399.85	56,059,832.41	62,413,898.70
EBITDA	\$	269,644,000.00	309,310,000.00	260,598,000.00	282,903,000.00	295,178,000.00	286,872,647.08	368,523,600.14	410,675,985.93	455,233,401.65	496,290,077.71
D&A as a % of EBITDA		15.58%	13.64%	16.65%	15.59%	15.79%	16.06%	12.44%	12.28%	12.31%	12.58%
Revenue	\$	553,525,000.00	616,619,000.00	576,429,000.00	647,684,000.00	650,611,000.00	676,105,892.20	772,794,831.26	843,410,914.34	916,628,355.89	986,527,459.42
D&A as % of Revenue		7.59%	6.84%	7.53%	6.81%	7.16%	6.81%	5.93%	5.98%	6.12%	6.33%
Depreciation as % of Gross PP&E		9.04%	9.52%	8.59%	7.09%	7.02%	7.25%	7.45%	7.65%	7.85%	8.05%
CapEx as % of Revenue		8.43%	6.37%	10.15%	7.11%	6.02%	7.05%	6.35%	6.97%	6.66%	6.44%
CapEx Coverage (CapEx / Depreciation)		2.33	1.58	2.13	1.78	1.38	1.455198721	1.314918023	1.372355977	1.253429052	1.152672158



Appendix - Comps

Ticker	Company Name	Current Price (\$)	Share Count	Market Capitalisation	Enterprise value	Revenue			EBITDA		Net Income	
						FY23	FY24	FY25	FY24	FY25	FY24	FY25
OLED	OLED TECHNOLOGIES	93.44	47,098,284.00	4,400,863,656.96	3,428,363,656.96	576,429,000.00	647,684,000.00	650,611,000.00	282,903,000.00	295,178,000.00	222,079,000.00	242,075,000.00
IDCC	InterDigital	355.13	25,700,000.00	9,126,841,000.00	8,340,841,000.00	549,590,000.00	868,520,000.00	834,000,000.00	496,200,000.00	538,400,000.00	358,610,000.00	406,600,000.00
DLB	Dolby Laboratories	59.98	95,500,000.00	5,728,090,000.00	5,124,090,000.00	1,299,000,000.00	1,274,000,000.00	1,349,000,000.00	368,000,000.00	368,000,000.00	262,000,000.00	255,000,000.00
ENTG	Entegris	107.98	151,900,000.00	16,402,162,000.00	19,911,162,000.00	3,524,000,000.00	3,241,000,000.00	3,197,000,000.00	919,000,000.00	919,000,000.00	181,000,000.00	293,000,000.00
VECO	Veeco Instruments	30.64	60,300,000.00	1,847,592,000.00	1,740,592,000.00	666,400,000.00	717,300,000.00	664,300,000.00	92,100,000.00	71,500,000.00	73,700,000.00	35,400,000.00
Mean		129.43	76,099,656.80	7,501,109,731.39	7,709,009,731.39	1,509,747,500.00	1,525,205,000.00	1,511,075,000.00	468,825,000.00	474,225,000.00	218,827,500.00	247,500,000.00
Median		93.44	60,300,000.00	5,728,090,000.00	5,124,090,000.00	982,700,000.00	1,071,260,000.00	1,091,500,000.00	432,100,000.00	453,200,000.00	221,500,000.00	274,000,000.00

Company Name	Revenue Growth		EBITDA Margin		Profit Margin	
	23'-24'	24'-25'	24'	25'	24'	25'
OLED TECHNOLOGIES	12.36%	0.45%	43.68%	45.37%	34.29%	37.21%
InterDigital	58.03%	-3.97%	57.13%	64.56%	41.29%	48.75%
Dolby Laboratories	-1.92%	5.89%	28.89%	27.28%	20.57%	18.90%
Entegris	-8.03%	-1.36%	28.36%	28.75%	5.58%	9.16%
Veeco Instruments	7.64%	-7.39%	12.84%	10.76%	10.27%	5.33%
Mean	13.93%	-1.71%	31.80%	32.84%	19.43%	20.54%
Median	2.86%	-2.67%	28.62%	28.01%	15.42%	14.03%

Implied Value Per Share	Equity Values		
	25' REVENUE	25' EBITDA	25' EPS
Minimum	\$ 2,675,754,223.56	\$ 4,399,393,656.96	\$ 4,400,863,656.96
Mean	\$ 4,654,742,457.82	\$ 6,537,063,031.71	\$ 8,291,626,575.72
Median	\$ 4,232,709,767.98	\$ 6,455,145,206.83	\$ 5,437,754,457.84
Maximum	\$ 7,477,796,071.76	\$ 8,156,826,718.55	\$ 13,551,376,676.28



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